

OS_Practical1

1. Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

Output:

```
aasritha@DESKTOP-4B676UM:~/OSlab-practical$ ./practical1
Enter a Linux command: uname

Parent Process
Parent PID : 65
Child PID  : 66

Child Process
Child PID : 66
Parent PID: 65
Linux

Child process completed.
aasritha@DESKTOP-4B676UM:~/OSlab-practical$ |
```

```
Enter a Linux command: ./practical1
```

Parent Process

Child Process

Parent PID : 67

Child PID : 68

Child PID : 68

Parent PID: 67

Architecture: x86_64

CPU op-mode(s): 32-bit, 64-bit

Address sizes: 39 bits physical, 48 bits virtual

Byte Order: Little Endian

CPU(s): 4

On-line CPU(s) list: 0-3

Vendor ID: GenuineIntel

Model name: Intel(R) Core(TM) i3-6006U CPU @ 2.00GHz

CPU family: 6

Model: 78

Thread(s) per core: 2

Core(s) per socket: 2

Socket(s): 1

Stepping: 3

BogoMIPS: 3983.99

Flags: fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc rep_good nopl xtopology cpuid pni pclmuldq vmx ssse3 fma cx16 pcid sse4_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch invpcid_single pti ssbd ibrs ibpb stibp tpr_shadow vnmi ept vpid ept_ad fsgsbase bmi1 avx2 smep bmi2 erms invpcid rdseed adx swap clflushopt xsaveopt xsavec xgetbv1 xsaves md_clear flush_l1d arch_capabilities

Virtualization features:

Virtualization: VT-x

Hypervisor vendor: Microsoft

Virtualization type: full

Caches (sum of all):

L1d: 64 KiB (2 instances)

L1i: 64 KiB (2 instances)

L2: 512 KiB (2 instances)

L3: 3 MiB (1 instance)

Vulnerabilities:

Itlb multihit: KVM: Mitigation: VMX disabled

L1tf: Mitigation: PTE Inversion; VMX conditional cache flushes, SMT vulnerable

Mds: Mitigation: Clear CPU buffers; SMT Host state unknown

Meltdown: Mitigation: PTI

Spec store bypass: Mitigation: Speculative Store Bypass disabled via prctl and seccomp

Spectre v1: Mitigation: usercopy/swapgs barriers and __user pointer sanitization

Spectre v2: Mitigation: Full generic retpoline, IBPB conditional, IBRS_FW, STIBP conditional, RSB filling

Srbds: Unknown: Dependent on hypervisor status

Tsx async abort: Not affected

Child process completed.

Child process completed.

```
aasritha@DESKTOP-4B676UM:~/OSlab-practical$ ./practical1
```

Enter a Linux command: lsblk

Parent Process

Parent PID : 69

Child PID : 70

Child Process

Child PID : 70

Parent PID: 69

NAME

	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINTS
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sda	8:0	0	256G	0	disk	
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sdb	8:16	0	256G	0	disk	/
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Child process completed.

Child process completed.

```
aasritha@DESKTOP-4B676UM:~/OSlab-practical$ ./practical1
```

Enter a Linux command: ps

Parent Process

Parent PID : 71

Child PID : 72

Child Process

Child PID : 72

Parent PID: 71

PID	TTY	TIME	CMD
9	pts/0	00:00:00	bash
71	pts/0	00:00:00	practical1
72	pts/0	00:00:00	ps

Child process completed.

```
aasritha@DESKTOP-4B676UM:~/OSlab-practical$ |
```

```
aasritha@DESKTOP-4B6700H: ~/703tab-practicals/7practical1
Enter a Linux command: top
```

```
Parent Process
Parent PID : 73
Child PID : 74
```

```
Child Process
Child PID : 74
Parent PID: 73
```

```
top - 05:54:08 up 56 min, 0 user, load average: 0.00, 0.00, 0.00
Tasks: 6 total, 1 running, 5 sleeping, 0 stopped, 0 zombie
%Cpu(s): 1.9 us, 3.8 sy, 0.0 ni, 94.2 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 3889.3 total, 3734.6 free, 221.1 used, 93.5 buff/cache
MiB Swap: 1024.0 total, 1024.0 free, 0.0 used, 3668.2 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	1804	1192	1104	S	0.0	0.0	0:00.08	init
7	root	20	0	1812	92	0	S	0.0	0.0	0:00.00	init
8	root	20	0	1812	100	0	S	0.0	0.0	0:00.44	init
9	aasritha	20	0	6068	5276	3552	S	0.0	0.1	0:00.26	bash
73	aasritha	20	0	2676	1080	988	S	0.0	0.0	0:00.00	practical1
74	aasritha	20	0	9320	5208	3060	R	0.0	0.1	0:00.04	top